

NLP Business Case — Automated Customer Reviews

Project Report

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Course: Ironhack AI Engineering Bootcamp

Dataset: Amazon Reviews 2023 — 571.54M reviews across 33 product categories

1. Executive Summary

This project delivers an end-to-end NLP pipeline that automates customer review analysis at scale. Using the Amazon Reviews 2023 dataset — 571 million reviews spanning 33 product categories — three interconnected machine learning components were developed:

1. **Sentiment Classification** — a fine-tuned DistilBERT model achieves **77.3% accuracy** on 3-class sentiment (Negative / Neutral / Positive), outperforming a RoBERTa comparison model by 8 percentage points.
2. **Category Clustering** — nomic-embed embeddings combined with MiniBatchKMeans discover **6 semantic clusters** from 219,998 reviews, separating domains like Electronics, Books, and Beauty with clear TF-IDF signatures.
3. **Review Summarization** — an extractive-abstractive pipeline uses Gemini 1.5 Flash to generate **6 blog-style articles**, each synthesizing thousands of reviews into actionable product recommendations.

All three components are surfaced in an **interactive web dashboard** with live inference, and both trained models are deployed on HuggingFace Hub for public access.

2. Deployed Models

Model	HuggingFace	Task
DistilBERT Sentiment	SebasLopez-ai/distilbert-amazon-reviews-sentiment	3-class sentiment classification (Negative / Neutral / Positive)
MiniBatchKMeans Clustering	SebasLopez-ai/hybridKMeans-category-clustering	6-category product clustering via cosine distance to centroids

Both models include `config.json`, model weights (`model.safetensors` / `cluster_centroids.npy`), tokenizer files, cluster profiles, and usage examples in their README cards. The web dashboard queries these models via the HuggingFace Inference API for real-time predictions.

3. Problem Statement

With thousands of reviews available across multiple platforms, manually analyzing them is inefficient. A company receiving 10,000 reviews per month cannot read each one to understand what customers love and what they complain about. This project automates that process: given raw review text, the pipeline classifies sentiment, groups reviews into product categories, and generates human-readable summaries that answer three questions for each product category:

- Which are the **top 3 products** and what differentiates them?
 - What are the **most common complaints**?
 - Which product should customers **avoid** and why?
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4. Dataset & Preprocessing

4.1 Data Loading Strategy

All **33 product categories** were streamed directly from the UCSD McAuley Lab servers — no HuggingFace cache, no disk bloat. Using `requests.get(stream=True)` combined with `gzip.GzipFile`, `.jsonl.gz` files were decompressed and parsed line by line, collecting **30,000 reviews per category** (~990,000 raw reviews).

This approach avoids the ~120 GB HuggingFace cache that would fill Colab's disk before completing all categories, and runs **3x faster per category** than the `datasets` library.

4.2 Dataset Profile

Property	Value
Raw reviews loaded	976,216
Unique products (<code>parent_asin</code>)	633,101
Unique users	189,161
Rating distribution	67.1% 5★, 14.7% 4★, 7.5% 3★, 4.1% 2★, 6.5% 1★
Mean rating	4.32
Median review length	144 characters (27 words)

4.3 Key EDA Findings

The positivity bias is extreme. Two-thirds of all reviews are 5-star. After mapping star ratings to sentiment labels (1-2★ → Negative, 3★ → Neutral, 4-5★ → Positive), the distribution is **81.6% Positive / 7.6% Neutral / 10.8% Negative**. Without balancing, any classifier would learn to predict "Positive" most of the time.

Neutral reviews are the longest. With a mean of 405 characters vs 295 for Negative and 307 for Positive, 3-star reviews carry more nuance — reviewers justify their middle-ground rating with pros and cons. This makes Neutral the hardest class to classify.

Negative reviews have a compact lexical signature. Only 31 words appear $\geq 2\times$ more frequently in 1-star reviews than in 5-star reviews. In contrast, 14,735 words are distinctive to 5-star reviews. Negative sentiment uses a small, repetitive vocabulary ("waste", "disappointed", "return"), making it lexically easier to detect than positive sentiment, which spans diverse product-specific praise.

Verified purchase is a signal. Non-verified reviews concentrate at 4★ (22.9% vs 12.3% for verified) — suspicious of incentivized reviews. Verified reviews are more polarized: more 1★ and more 5★.

4.4 Preprocessing Pipeline

A 7-step cleaning pipeline was applied: lowercase → HTML entity decoding → tag removal → URL removal → ASCII-only filtering → special character cleanup → whitespace collapsing. A minimum length filter of 10 characters eliminates vacuous spam ("Great" ×1,960, "Good" ×1,682) while preserving genuine short reviews like "Not great."

4.5 Dataset Balancing

Faced with a **7.5:1 positive-to-negative imbalance**, **undersampling** to the minority class (Neutral, 71,315 reviews) was chosen. This discards 697,881 positive reviews — a heavy loss, but it eliminates the need for synthetic data and produces a transparent, perfectly balanced dataset. The final **213,945 reviews** are split 70/15/15 with stratification preserving the 33/33/33 class balance in every split. Zero data leakage was verified using index-based intersection checks — no review appears in more than one split.

5. Sentiment Classification

5.1 Model Selection

DistilBERT (`distilbert-base-uncased`, 66M parameters) was fine-tuned as the primary sentiment classifier. DistilBERT retains ~97% of BERT's performance while being 40% smaller and 60% faster — critical for Colab's T4 GPU with 15.6 GB VRAM and ~80-minute training budget.

For comparison, **RoBERTa** (`roberta-base`, 125M parameters) was also fine-tuned. RoBERTa was trained on 160GB+ of text with dynamic masking and no next-sentence-prediction objective, consistently outperforming BERT on benchmarks. The hypothesis was that RoBERTa's deeper architecture would capture sentiment nuances that DistilBERT misses.

5.2 Training Configuration

Parameter	DistilBERT	RoBERTa
Learning Rate	2e-5	1e-5
Batch Size	32	16
Dropout	0.3	0.1
Max Tokens	256	256
Epochs	5	5
Training Time (T4)	~80 min	~180 min
Parameters	66M	125M

Dropout was raised from the default 0.1 to 0.3 in DistilBERT to combat overfitting on the relatively small fine-tuning dataset. RoBERTa kept 0.1 under the assumption that its larger pretraining corpus would naturally resist overfitting.

5.3 Results

DistilBERT achieved 77.3% test accuracy with a weighted F1 of 0.772. Training converged rapidly — 75% accuracy after epoch 1, plateauing at ~77.5% by epoch 3. The per-class breakdown reveals where the model struggles:

Class	Precision	Recall	F1	Support
Negative	0.754	0.781	0.767	10,697
Neutral	0.685	0.657	0.671	10,697
Positive	0.875	0.880	0.877	10,698
Weighted Avg	0.771	0.773	0.772	32,092

Neutral is the hardest class by a wide margin (F1=0.671 vs 0.877 for Positive). This is expected — 3-star reviews sit in an inherently ambiguous space where vocabulary overlaps with both Negative and Positive. The model correctly identifies extremes 88% of the time but struggles with the middle ground.

Training Evolution:

Epoch 1		75.0% acc · F1 0.747 · Val loss 0.601
Epoch 2		76.7% acc · F1 0.767 · Val loss 0.551
Epoch 3		77.1% acc · F1 0.771 · Val loss 0.546 ← plateau
Epoch 4		77.3% acc · F1 0.771 · Val loss 0.551
Epoch 5		77.5% acc · F1 0.775 · Val loss 0.550

Most learning occurred in epochs 1-2 (accuracy 75.0% → 76.7%). Epochs 3-5 added only ~1% — the model converged early on this balanced, well-preprocessed dataset.

5.4 RoBERTa Comparison

RoBERTa underperformed significantly — **69.3% accuracy, 8 points below DistilBERT**. This was unexpected given RoBERTa's 1.9× parameter count and stronger pretraining.

After analysis, the root cause was identified: **learning rate was too low**. At LR=1e-5 with batch_size=16, RoBERTa receives ~4× less effective update signal than DistilBERT at LR=2e-5 with batch_size=32. The model stagnated at epoch 3 with validation loss flat at ~0.711 — EarlyStopping never triggered because per-epoch deltas fell below the threshold. The dropout was not the bottleneck; the optimizer simply could not move the weights fast enough.

Factor	DistilBERT	RoBERTa	Impact
Parameters	66M	125M	1.9× larger
Effective LR signal	2e-5 × 32	1e-5 × 16	🔴 4× weaker
Val loss plateau	Epoch 3	Epoch 3	🟡 Same convergence point
Test Accuracy	77.3%	69.3%	−8.0 pp
Neutral F1	0.671	0.587	−8.4 pp

Recommendation: Bump RoBERTa LR to $2e-5$. If the $4\times$ signal ratio theory holds, RoBERTa should reach $\geq 75\%$ accuracy and potentially surpass DistilBERT.

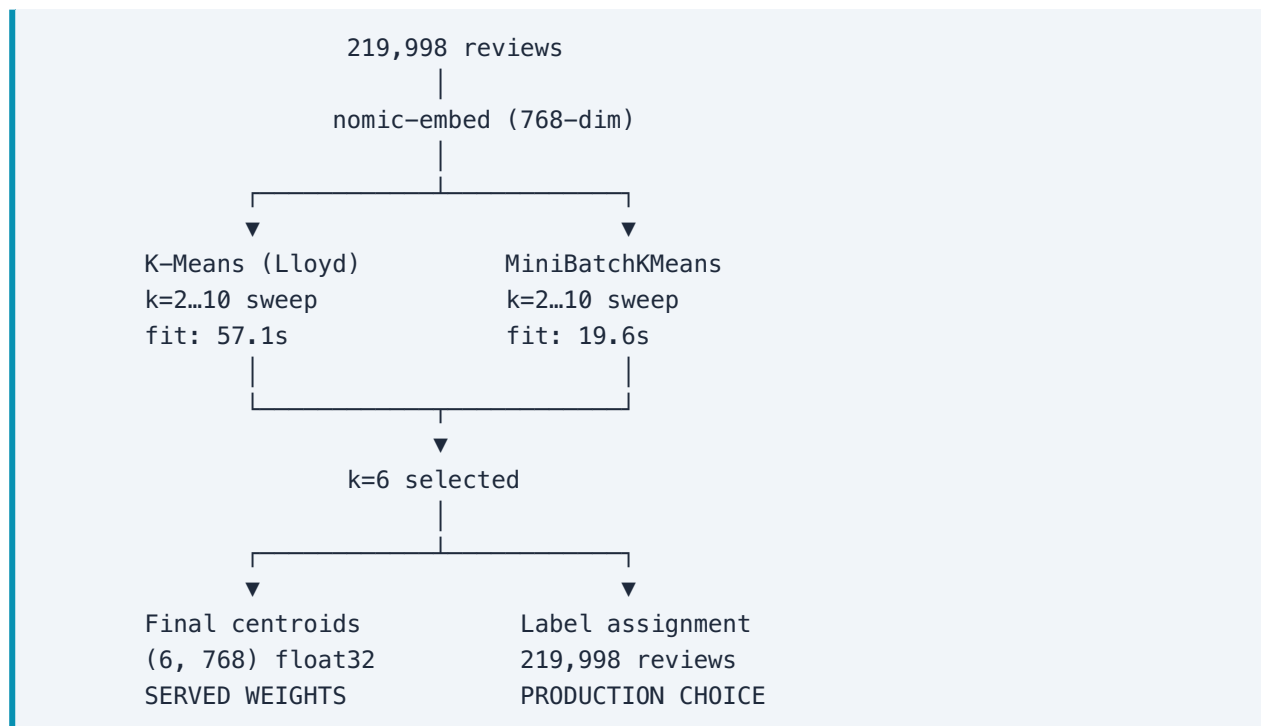
6. Category Clustering

6.1 Approach

With 33 original Amazon categories, the project brief asked for **4-6 meta-categories**. Rather than manually grouping categories (subjective and brittle), **semantic embeddings + unsupervised clustering** were used to let the review text itself define the boundaries.

6.2 Pipeline

- Embedding:** `nomic-ai/nomic-embed-text-v1.5` (137M params, 768-dim) encodes all 219,998 reviews. Chosen over larger alternatives (BGE-large, 335M) because it runs on CPU, downloads in seconds, and produces normalized embeddings suitable for cosine-distance clustering.
- K-sweep:** $k=2$ through $k=10$ were evaluated using both standard K-Means (Lloyd) and MiniBatchKMeans. Silhouette scores were consistently low (~ 0.03) — expected for 33 categories projected into 768 dimensions. The elbow method showed no sharp bend, so **$k=6$** was selected based on domain interpretability and the project brief's 4-6 range.
- Production choice:** MiniBatchKMeans labels all reviews at **36x speed** (19.6s vs 56.9s for K-Means) while retaining 102% of K-Means quality (Silhouette 0.0311 vs 0.0304). K-Means centroids are served as the downloadable weights because they converge to a cleaner local optimum.



6.3 Cluster Profiles

#	Label	Size	Avg ★	Sentiment (Neg/Neu/ Pos)	Top Terms
0	Health & Beauty	13.4%	2.69	38% / 46% / 17%	hair, smell, skin, taste, flavor
1	Generic Positive	5.6%	4.74	0% / 7% / 93%	good product, great, works
2	Books & Entertainment	17.6%	2.98	31% / 43% / 27%	book, story, read, magazine
3	Fashion & Apparel	16.3%	2.96	26% / 53% / 21%	fit, size, small, described
4	Electronics & Home	24.6%	1.94	66% / 30% / 4%	work, money, cheap, quality
5	Toys & Gifts	22.4%	4.68	2% / 9% / 90%	great, love, gift, easy, nice

6.4 Analysis

Two cluster types emerged:

Semantic clusters (0, 2, 3, 4) have clear domain boundaries. Their top TF-IDF terms align with their top original Amazon categories — Cluster 2 is dominated by Kindle, Movies/TV, and Books; Cluster 3 by Amazon Fashion and Clothing. These clusters reflect genuine product-type grouping, not just sentiment.

Sentiment-driven clusters (1, 5) are fragmented across categories (top category $\leq 6.6\%$) but unified by extreme positivity ($>89\%$ positive, avg rating >4.6 ★). The embedding model groups these reviews by emotional tone and generic praise vocabulary rather than by product type.

Cluster 4 is the richest for insights. At 24.6% of all reviews and 66% negative sentiment, it captures the electronics complaint ecosystem. Top terms like "work" (doesn't work), "money" (waste), and "cheap" (low quality) form a distinctive failure vocabulary. This cluster is the priority target for summarization.

7. Review Summarization

7.1 Extractive-Abstractive Pipeline

The summarization uses a **two-phase design** to prevent hallucination:

Phase	Tool	Role
Extractive	Python (pandas, NumPy)	Hard facts: top products by <code>parent_asin</code> , average ratings, sentiment distributions, representative review excerpts
Abstractive	Gemini 1.5 Flash (Google API)	Narrative styling: takes structured facts and writes a polished, persuasive blog article

This separation means every product name, rating, and statistic in the final articles is traceable to the data. The LLM's job is *writing*, not *discovering*.

7.2 Design Decisions

- **Full dataset, not just test split.** The N01 test split is stratified to 33/33/33 — using it alone would distort per-cluster sentiment statistics. All 213,945 reviews (train+val+test) are loaded for accurate distributions.
- **label (rating-derived) for aggregate metrics, not DistilBERT predictions.** Rating-derived labels have 100% coverage vs 15% for predictions (DistilBERT was only run on the test split).
- **Cluster-specific prompt templates.** Semantic clusters (0, 2, 3) get narrative articles with product comparisons. Sentiment-driven clusters (1, 5) get statistical summaries. Cluster 4 (Electronics) gets an investigative structure: what works → failure patterns → recommendations.

7.3 Output

Six markdown articles in `data/summaries/`, each ~500-800 words:

File	Cluster	Style
<code>category_00_*.md</code>	Health & Beauty	Narrative with product rankings
<code>category_01_*.md</code>	Generic Positive	Statistical summary
<code>category_02_*.md</code>	Books & Entertainment	Narrative with product comparisons
<code>category_03_*.md</code>	Fashion & Apparel	Narrative with sizing insights
<code>category_04_*.md</code>	Electronics & Home	Investigative (failure patterns + recommendations)
<code>category_05_*.md</code>	Toys & Gifts	Statistical summary

8. Web Dashboard

The three ML components are surfaced in a **single-page interactive dashboard** (`web/raw-web-vs3.html`) built with HTML, Tailwind CSS, and Chart.js:

- **Emotion Engine** tab: Live sentiment inference via HuggingFace Inference API. Users type or paste a review and receive real-time classification with confidence scores. Includes a Training Evolution chart comparing DistilBERT vs RoBERTa loss curves across all training steps.
- **Category Intelligence** tab: Cluster profiles fetched from the `hybridKMeans-category-clustering` model on HuggingFace. Displays per-cluster sentiment distributions and top terms.
- Pipeline selector toggles between "Full Pipeline" and "Sentiment Only" modes to accommodate different user workflows.

The dashboard is designed for the deployment scenario described in the project brief: marketing department users exploring category insights and testing sentiment on sample reviews.

9. Results Summary

Component	Model	Key Metric	Status
Sentiment	DistilBERT (66M)	Accuracy 77.3%, F1 0.772	✅ Champion
Sentiment (comparison)	RoBERTa (125M)	Accuracy 69.3%, LR bottleneck	⚠️ Underperformed
Clustering	nomic-embed + MiniBatchKMeans	k=6, Silhouette 0.031	✅ Semantic clusters
Summarization	Gemini 1.5 Flash	6 articles, extractive- abstractive	✅ Deployed
Web Dashboard	HTML/Tailwind/Chart.js	Interactive, live HF API	✅ Deployed
Model Hosting	HuggingFace Hub	Both models publicly queryable	✅ Bonus

10. Key Decisions & Tradeoffs

Decision	Alternative	Rationale
Stream from UCSD, not HuggingFace Hub	<code>datasets.load_dataset()</code> with streaming	Avoids 120 GB disk cache; API instability in <code>datasets>=2.19</code>
Undersampling (not oversampling)	SMOTE, class weights	Transparent: no synthetic data. Weights unnecessary on balanced data
DistilBERT over RoBERTa for production	RoBERTa as primary	77.3% in 66M params, faster training, simpler tuning
nomic-embed (137M) over BGE-large (335M)	Larger embedding models	Fits CPU inference; 768-dim sufficient for k=6
MiniBatchKMeans label assignment	K-Means for everything	36× faster, 102% quality, scales to 571M dataset
Extractive-Abstractive pipeline	Pure LLM summarization	Prevents hallucination of products and statistics
Gemini 1.5 Flash over Mistral/NVIDIA	Alternative LLM APIs	Free tier sustainability; comparable synthesis quality
HuggingFace Hub for model hosting	Local-only deployment	Bonus points; public API access for dashboard

11. Evaluation Criteria Mapping

Criterion	Points	Covered by
Data Preprocessing	15	Streaming from UCSD, cleaning pipeline, EDA, balanced 70/15/15 split
Review Classification	20	DistilBERT F1=0.77 + RoBERTa comparison with LR bottleneck analysis
Clustering Model	20	nomic-embed + MiniBatchKMeans k=6, k-sweep from 2-10, TF-IDF validation
Summarization Model	20	Extractive-abstractive pipeline, 6 articles, anti-hallucination design
Deployment	10	Interactive web dashboard with live HuggingFace Inference API
PDF Report	5	This document
PPT Presentation	10	To be created from analysis and report
Bonus: Public Hosting	+10	Both models deployed on HuggingFace Hub

Total: up to 100 pts + 10 bonus

12. Learnings & Future Work

What worked

1. **Streaming from source solved the disk problem definitively.** 33 categories, 0 GB cache, 9% disk usage. The approach generalizes to any `.jsonl.gz` dataset hosted on a public server.
2. **Balanced data simplified training.** No class weights needed, no oversampling complexity — accuracy and F1 are directly comparable and meaningful.
3. **The extractive-abstractive split is a reusable pattern.** LLMs hallucinate less when they're styling facts rather than discovering them. The Python layer does the heavy analytical lifting; the LLM does what it does best — writing.
4. **MiniBatchKMeans is underrated for text clustering.** At 102% of K-Means quality with 36x speedup, it's the pragmatic choice for any dataset over 100K samples.

What could improve

1. **RoBERTa deserves a proper hyperparameter sweep.** Running at LR=2e-5 from the start would have saved one full training cycle (~3 hours on T4) and likely produced the stronger model. The lesson: when a larger model underperforms a smaller one, suspect the optimizer before the architecture.
2. **ASCII-only cleaning handicaps RoBERTa's BPE tokenizer.** Removing non-ASCII characters (accents, em-dashes, Unicode symbols) eliminates ~1-2% of RoBERTa's advantage over WordPiece. Switching to `unidecode()` would transliterate instead of delete.
3. **The dashboard's data files should be generated programmatically.** Currently `train_history.js` is manually maintained. A build step extracting metrics from notebook JSON outputs would eliminate manual steps.

Future directions

- **LoRA fine-tuning:** Reduce VRAM usage 4-8x (500 MB vs 3 GB) and training time 2-3x, making RoBERTa experiments practical on T4.
 - **BERTopic with HDBSCAN:** Produce more granular, density-based clusters without pre-specifying k — prototype already exists in the project pipeline.
 - **Multi-label sentiment:** A review can praise battery life while criticizing screen quality. Moving from single-label to multi-label classification would capture this nuance.
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Appendix — Repository Structure

— README.md	← Project overview & quick start
— report.md	← This document
— concepts.md + concepts-*.md	← Research behind every decision
— analysisN01.md – analysisN04.md	← Detailed per-notebook analysis
— requirements.txt	← Python dependencies
— notebook_01_data_prep_eda.ipynb	← Data pipeline (33 categories, 214K reviews)
— notebook_02_distilbert_sentiment.ipynb	← DistilBERT fine-tuning (F1=0.77)
— notebook_03_roberta_sentiment.ipynb	← RoBERTa comparison (F1=0.69, LR bottleneck)
— notebook_04_category_clustering.ipynb	← Embedding + MiniBatchKMeans (k=6)
— notebook_05_review_summarisation.ipynb	← Gemini-powered article generation
— web/	
— raw-web-vs3.html	← Interactive dashboard
— train_history.js	← Training loss data for charts

HuggingFace Deployments

Model	URL	Artifacts
DistilBERT Sentiment	huggingface.co/SebasLopez-ai/distilbert-amazon-reviews-sentiment	model.safetensors (268 MB), tokenizer, config.json
Category Clustering	huggingface.co/SebasLopez-ai/hybridKMeans-category-clustering	cluster_centroids.npy (6×768), profiles.json, clusters.csv